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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Ctrl + Future
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 – Decent Work and Economic Growth
Project / Solution Name	SME Smart
Name (Leader)	Teh Sze Yu Jocelyn
Email & Phone (Leader)	Jocelynteh0602@gmail.com / 0129202655
University Affiliation(s)	University Technology Malaysia (UTM) / University Putra Malaysia (UPM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Many small and medium-sized enterprises in Malaysia still rely on processes and disconnected applications to manage daily business operations, including customer enquiries order processing, invoicing, appointment scheduling and email communication.

These repetitive administrative tasks consume time increase the risk of human errors and often lead to delayed customer responses and inefficient workflows. Small businesses with manpower and technical expertise face additional challenges in adopting digital solutions as existing systems may be costly, complicated or difficult to customize according to their business needs. As businesses grow managing information, across platforms becomes increasingly challenging affecting employee productivity, customer satisfaction and overall business efficiency.

This issue directly relates to SDG 8 – Decent Work and Economic Growth which promotes productivity, innovation and sustainable business development. Therefore SMEs need a accessible and adaptable solution that reduces repetitive workloads connects essential business processes and enables employees to focus on higher-value activities.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SME Smart is an AI-powered platform designed to help small and medium-sized enterprises (SMEs) manage their daily tasks more efficiently. Instead of offering the same AI solution to every business, we wanted to let SMEs customize their own AI assistants based on their needs.

To make this possible, we use Qwen2.5-7B-Instruct with vLLM on RunPod to generate customized Standard Operating Procedure (SOP) prompts. These prompts are then used in Flowise to set up AI workflows for different businesses.

One of our main features is an AI chatbot that can answer customer enquiries, take orders, and generate invoices automatically. Employees still handle important decisions, such as verifying payments, updating invoice statuses, and arranging deliveries. We also included Excel product imports, appointment scheduling with Google Calendar, AI-assisted Gmail management, and an internal AI assistant that helps staff access business information.

We built SME Smart using HTML, CSS, JavaScript, PHP, Python, and PostgreSQL, with Replit supporting our development and deployment. Throughout the project, we learned how to integrate different technologies and overcome challenges in building a working AI system.

By bringing these features together, we hope to reduce repetitive work, save time, and help SMEs improve productivity. We also plan to explore possible integration with the Orbyte ecosystem in the future. Overall, SME Smart supports SDG 8 by helping small businesses grow through technology.

3. Technical Architecture & Implementation Plan

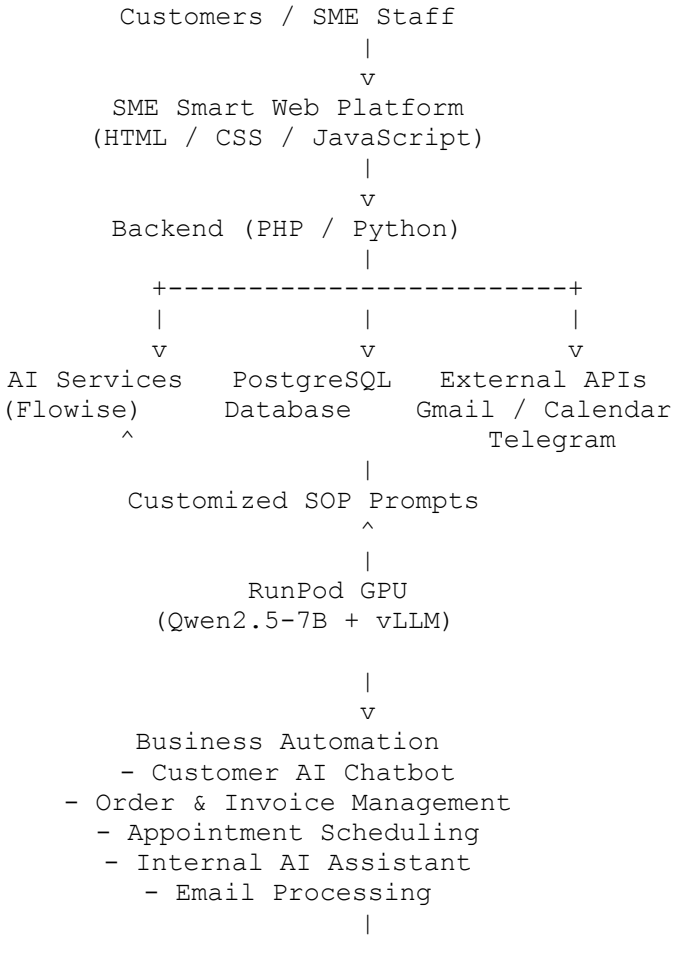
3.1 Conceptual Tech Stack

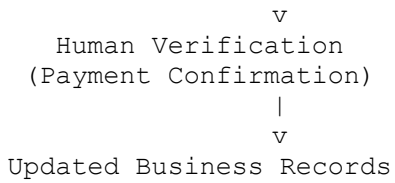
List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

SME Smart uses HTML, CSS, and JavaScript for frontend development, PHP and Python for backend operations, and PostgreSQL for database management. Flowise manages AI chatbot workflows, while Qwen2.5-7B-Instruct, deployed through vLLM on RunPod, generates customized SOP prompts for different businesses. Gmail API and Google Calendar API support email management and appointment scheduling. Replit is used for web application development and deployment.

System Architecture





Yes. Our project utilizes Replit for web application development, testing, and cloud deployment. We also use RunPod to host the Qwen2.5-7B-Instruct model with vLLM, generating customized SOP prompts based on each company's business requirements. These prompts are integrated into Flowise to configure personalized AI assistants.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We started by looking at the everyday challenges faced by SMEs, such as repetitive manual work, disconnected systems, and slow responses to customers. From there, we came up with SME Smart, an AI-powered solution designed to help small businesses save time and work more efficiently.

During the planning and development stages, we focused on building useful features such as customizable AI assistants, automated order and invoice management, and appointment scheduling. We also wanted to keep the dashboard simple and easy to use, while allowing employees to stay in control of important decisions.

To bring our idea to life, we used PHP, Python, PostgreSQL, Flowise, and RunPod. Along the way, we faced challenges with customizing AI responses, API authentication, and connecting different parts of the system. By dividing tasks based on our team members' strengths and working closely together, we managed to build a working prototype within the limited time.

For Phase 2, we plan to test SME Smart with real SMEs, improve AI accuracy, strengthen data security, and make the system more reliable. We also hope to improve AI personalization and explore how SME Smart could work with the Orbyte ecosystem.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.2 - Achieve higher levels of economic productivity through technological upgrading and innovation.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Based on our initial testing, manually responding to five customers and recording their orders in Excel took 20 minutes. Including invoice preparation, we estimate the total process would take around 30 minutes.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage reduction in the time needed to process five customer orders and generate invoices compared to manual processing.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	We aim to reduce order-processing and invoicing time by at least 80%, with SME Smart handling five customer orders and generating invoices within 3 minutes.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] Jocelyn

[Date] 24/10/2026